Jack Agterdenbos

MSc Aerospace Engineering · Sustainable Air Transport · AI & Mathematical Optimisation

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PROFILE

Aerospace engineer with an MSc in Sustainable Air Transport from TU Delft (completed October 2026), specialising in AI-driven trajectory optimisation and autonomous systems. Experience of end-to-end technical delivery: built a ULD availability forecasting tool at KLM Cargo (deployed in its planning environment) and sole-developed a hardware prototype for Juvoly's clinical AI pipeline. Combines a solid aerospace foundation with applied machine learning and reinforcement learning.

SKILLS

Programming: Python, C++, Git

AI & Machine Learning: Reinforcement Learning, Goal-Conditioned RL, Supervised & Unsupervised Learning, Surrogate Modelling

Simulation & Modelling: Trajectory Modelling, Operations Research, Data Pipelines, Predictive Analytics

Domain Knowledge: Aerospace Systems, Sustainable Aviation, Airline Cargo Operations, Embedded Systems (ESP32)

EXPERIENCE

Intern, Cargo Control Center · KLM Cargo · *Sep 2025 – Jan 2026*

- Engineered an end-to-end ULD (Unit Load Device) availability forecasting system in Python, building the full stack from raw data ingestion and pipeline design through to a scikit-learn predictive model deployed directly in an operational cargo planning environment.
- Integrated the forecasting system into cargo control workflows, providing cargo control planners with ULD availability predictions.

Student Prototype Developer · Juvoly · *Jul 2025 – Present*

- Sole developer of a portable wireless audio recorder for clinical consultations, engineering the firmware in C++ on an ESP32 microcontroller with AES encryption, secure local storage, and seamless integration with Juvoly's AI transcription pipeline.
- Managed the full development cycle independently from concept to working prototype, intended for Juvoly's platform spanning 50+ medical specialisms in GP, hospital, and occupational health settings.

Private Tutor, Maths, Physics & Chemistry · DeBijlesStudent B.V. · *Nov 2021 – Present*

- Delivered one-to-one instruction at secondary and pre-university level.

Earlier Roles *Front Office Employee, Speelpark Oud Valkeveen (Jul–Aug 2022) · Grocery Delivery Driver, Picnic Technologies (Jun–Aug 2021) · Stock Clerk, Albert Heijn (May 2017–Dec 2020)*

EDUCATION

MSc Aerospace Engineering, Sustainable Air Transport · TU Delft · *Sep 2024 – Oct 2026*

Thesis: Trajectory-Based Aircraft Landing Problem (TBALP) Formalises the TBALP under a Goal-Conditioned Hierarchical Reinforcement Learning (GCHRL) framework, implemented in Python using bluesky-gym and Stable-Baselines3. A classical planner (Constrained Position Shifting) assigns runways and arrival times, supported by an Extra Trees surrogate for fast arrival-time estimates; a Soft Actor-Critic agent adapts flight path execution to meet those targets.

BSc Aerospace Engineering · TU Delft · *Sep 2021 – Jul 2024*

Semester Exchange, Business Administration · Universidad Carlos III de Madrid · *Sep 2023 – Jan 2024*

Broadened commercial and strategic perspective alongside core aerospace training; strengthened cross-cultural collaboration in a multilingual environment.

VWO Pre-University Education · Vituscollege · *Sep 2019 – Jun 2021*

LANGUAGES

Dutch: Native · **English:** Native / Bilingual · **German:** Limited Working Proficiency · **Spanish:** Elementary